Bursty infectiousness as a hidden driver of epidemic variability

Stephen M. Kissler[1,3,4]* [Full authorship and ordering TBD]

**Corresponding author: stephen.kissler@colorado.edu*

Abstract

Epidemic trajectories are profoundly shaped by random, individual-level effects. Superspreading can make epidemics less likely to establish, but more explosive when they do; and chance events during the cryptic early stages of an outbreak can durably shift its overall timing.[1, 2] Accounting for such effects is vital for public health preparedness: while a good understanding of an epidemic's expected trajectory can be achieved using deterministic, population-level models, the variation around that mean — including best- and worst-case scenarios — hinges upon individual variation in transmission capacity and timing.

One such factor that has so far received little attention is the concentration of individual infectiousness, or its "burstiness". Evidence from SARS-CoV-2 and influenza suggests that the window of meaningful infectiousness for those viruses may be short — even less than a day — despite the fact that the infectiousness window is typically quoted as lasting 6-8 days for influenza and 9-12 days for SARS-CoV-2.[3, 4, 5] These longer spans reflect the generation interval distribution, $g(\tau)$, a population-level average of the times τ between primary and secondary infections. Its individual-level counterpart, the infectiousness kernel[6, 7], describes the distribution of transmission times for a single infector, and need not resemble $g(\tau)$: $g(\tau)$ can be broad simply because different individuals' infectiousness windows fall at different times, even if each individual is only briefly infectious. If the infectiousness kernel is in reality much more concentrated than $g(\tau)$, this could impact everything from how individuals gauge their own risk of transmitting to how epidemics unfold overall.

Although individual-based simulations can in principle accommodate heterogeneous infectiousness timing, standard analytical frameworks operate on the population-level $g(\tau)$ directly and thus are blind to burstiness by construction. More fundamentally, no existing framework has been designed to isolate burstiness as a free parameter while holding $g(\tau)$ and R_0 fixed, making it impossible to attribute observed epidemic dynamics specifically to burstiness. Empirical evidence is likewise limited: resolving the infectiousness kernel requires densely sampled, individual-level transmission data that is rarely available, and what data exist are subject to biases that can both inflate and obscure the apparent degree of burstiness. Whether burstiness has epidemiologically meaningful consequences therefore remains an open question.

Here, we address that gap by systematically examining the impact of bursty infectiousness on epidemic dynamics. We introduce the burstiness parameter, $\psi \in [0, 1]$, which governs the width of the individual infectiousness kernel — specifically, the fraction of $g(\tau)$'s variance attributable to within-infector variation in transmission timing (**Supplementary Methods**). At the $\psi = 0$ extreme (maximum burstiness), each infector transmits at a single moment, with that moment drawn from $g(\tau)$. At the $\psi = 1$ extreme, each infector's secondary infections are distributed according to $g(\tau)$ itself. Critically, varying ψ has no impact on the fundamental population-level epidemiological quantities, $g(\tau)$ and R_0. In this respect, ψ stands in the same relation to $g(\tau)$ as the overdispersion parameter k stands to R_0 [1]: a measure of individual-level variation that is invisible to population-level averages yet shapes epidemic dynamics. We show that burstiness drives an underappreciated form of superspreading, inflates variability in outbreak timing, shapes how isolation and gathering-size restrictions affect epidemic dynamics, and complicates estimation of key epidemiological parameters.

To embed ψ within a framework suitable for analyzing its epidemiological properties, we introduce the Gamma burst model. The model assumes $g(\tau)$ is (approximately) Gamma-distributed with shape α and rate β. At the individual level, the infectiousness kernel is a Gamma$(\psi\alpha, \beta)$ distribution — the "burst" — following a latent period of length $l \sim$ Gamma$((1-\psi)\alpha, \beta)$ (**Figure XX**). Since both components are Gamma-distributed with the same rate, their sum recovers Gamma$(\alpha, \beta) = g(\tau)$, ensuring that any value of ψ yields the same population-level generation interval distribution. The Gamma burst model has biological grounding in within-host viral dynamics: stochastic viral replication early in infection creates between-individual variation in the onset of detectable viral growth (the variable latent period), while subsequent shedding kinetics are more consistent across individuals (the infectiousness burst)[8].

Empirical estimates of ψ. To gauge the potential range of burstiness across pathogens, we estimated ψ using published transmission trees with serial intervals for COVID-19, MERS, measles, Ebola, smallpox, pneumonic plague, norovirus, and hepatitis A [9] (**Methods**). The resulting estimates spanned a wide range (**Figure XX**): measles and pneumonic plague showed the strongest evidence of burstiness (95% credible interval for ψ: [0.00, 0.11] and [0.00, 0.20], respectively), while smallpox and MERS showed evidence of more diffuse infectiousness (95% credible interval for ψ: [xx,xx] and [xx,xx], respectively).

These estimates should be regarded with caution. Multiple biases can distort estimates of ψ in either direction: uncertainty in infection timing can artificially disperse observed infection times, biasing ψ upward; whereas an emphasis on recording superspreading events while ignoring less explosive transmission could bias ψ downward. Moreover, since $g(\tau)$'s variance must reflect at least some between-individual variation in infectiousness timing, values of ψ near 1 are likely overestimates. Nevertheless, the range of estimates — along with independent viral kinetics evidence suggesting narrow infectiousness windows for influenza and SARS-CoV-2 [3] — indicates that ψ varies across pathogens, and that bursty infectiousness reflects a genuine epidemiological feature rather than a measurement artifact.

For the remainder of our analyses, we illustrate our main findings using three benchmark pathogens spanning a range of reproduction numbers and generation interval shapes (**Figure XX**): influenza ($R_0 = 2$), SARS-CoV-2 omicron ($R_0 = 6$), and measles ($R_0 = 12$). For each, we examine the impact of burstiness across its full range ($\psi \in [0, 1]$), though the methods and findings are general.

Bursty infectiousness drives coincidence superspreading. Superspreading happens when a person infects significantly more than the expected number of secondary cases (R_0). Typically, superspreading is assumed to arise from persistent individual-level traits: either a person has more than the average number of contacts, or sheds more than the typical amount of pathogen. The mechanism matters: contact-driven and infectiousness-driven superspreading produce qualitatively different epidemic dynamics and require different types of response.[10]

A third type of superspreading can arise purely by chance, when an infectiousness burst happens to align with a brief period of high contacts. This "coincidence superspreading" is fundamentally different from other types: it does not arise from any persistent individual trait, and therefore cannot be targeted by individual-level interventions. Coincidence superspreading thus represents a floor on overdispersion that persists even after all contact-driven and infectiousness-driven heterogeneity has been accounted for.

Bursty infectiousness leads to more coincidence superspreading: a burst that happens to coincide with a high-contact period can generate a superspreading event, while broad infectiousness kernels tend to average over contact variation and make superspreading less likely. This can be illustrated by considering a simple periodic model for contact rates, $c(t) = 1 - \rho\cos(2\pi t/\omega)$, where $\rho \in [0,1]$ is the amplitude and $2\pi/\omega$ is the period of the contact variation, relative to a baseline contact rate of $\bar{c} = 1$ (**Supplementary Methods**). Crucially, this contact model describes contact variation over time but not across individuals; each person is assumed to be identical in every way except for when their infectiousness window falls, so any overdispersion that emerges reflects coincidence superspreading alone. Under the Gamma burst model, and assuming that infections occur at uniform-random times relative to the contact process, we can calculate the overdispersion (OD) in secondary infection counts in terms of the Lloyd-Smith parameter, k:

$$\mathrm{OD} \equiv 1/k = \frac{\rho^2}{2}\left(1 + \left(\frac{2\pi\bar{g}}{\alpha\omega}\right)^2\right)^{-\psi\alpha} \tag{1}$$

Here, α is the shape parameter of the generation interval distribution and $\bar{g} = \alpha/\beta$ is the mean generation time. Small values of k correspond to high overdispersion, while $k \to \infty$ denotes a Poisson distribution of secondary cases (mean equal to variance, no overdispersion).

OD is maximized by bursty infectiousness ($\psi \to 0$), with its sensitivity to ψ governed by the ratio of the contact variation period ω to the mean generation time $\bar{g}$. Burstiness has the greatest impact on OD when these two timescales are comparable ($2\pi\bar{g}/(\alpha\omega) \approx 1$); faster contact variation gets averaged over by any infectiousness kernel and slower variation produces overdispersion regardless of ψ (**Supplementary Methods**). For the benchmark pathogens, where $\bar{g}$ ranges from roughly 3–12 days, weekly contact variation falls squarely in this sensitive zone. Empirical contact surveys confirm that contacts vary meaningfully on this timescale: the POLYMOD and BBC Pandemic studies document 20%–40% more contacts on weekdays than weekends, corresponding to $\rho \approx 0.1$–0.2 at $\omega = 7$ days.

For the three benchmark pathogens (influenza, SARS-CoV-2 omicron, and measles), simulations confirm these analytic predictions across a range of contact amplitudes and periods. The resulting overdispersion has direct consequences for outbreak establishment, with burstier infectiousness translating into higher epidemic extinction probabilities (**Figure XX**). Similar patterns emerged for stochastic contact processes, suggesting robustness to the specific form of contact variation (**Supplementary Methods, Supplementary Figure XX**).

Burstiness and the epidemic latent period. Like infections, epidemics also have a "latent period" — the span between the introduction of the first case and when the outbreak becomes established. Early-outbreak stochasticity shapes this latent period's length, creating variability in epidemic timing even among identically parameterized outbreaks.[2] Bursty infectiousness amplifies this variability, producing outbreaks that are more erratic in timing and, on average, slower to establish.

To quantify this, we express the early epidemic as a branching process $Z(t) \approx We^{rt}$, where W is a random initial condition capturing early-outbreak stochasticity — equivalently expressible as a time shift ς, so that $Z(t) \approx e^{r(t+\varsigma)}$ [2]. Bursty infectiousness inflates $\mathrm{Var}[W]$, which makes epidemic timing more variable (increases $\mathrm{Var}[\varsigma]$) and shifts the typical epidemic slightly later (decreases $E[\varsigma]$). Formally,

$$\mathrm{Var}[W] \propto z(R_0, \alpha)^{(1-\psi)\alpha} \tag{2}$$

where z is a quantity that depends on R_0 and α; it is increasing in R_0 and is greater than 1 when $R_0 > 1$. Thus, bursty infectiousness ($\psi \to 0$) maximizes $\mathrm{Var}[W]$, with the largest effect when R_0 is high. If, following Morris et al. [2], we assume W (conditional on epidemic survival) is approximately Gamma-distributed, this translates into more variable and later-shifted epidemic timing (**Supplementary Methods**).

Simulations confirm these predictions across the benchmark pathogens (**Figure XX**). Survival curves depicting the time to reach 5% total prevalence show marked differences across ψ-values for SARS-CoV-2 omicron and measles; whereas differences for influenza, with its lower R_0, were negligible. For measles, xx% of simulated outbreaks reached 5% prevalence by week 5 at the smooth extreme ($\psi = 1$), compared with only xx% at the bursty extreme ($\psi = 0$), consistent with bursty infectiousness mattering most at high R_0.

Common interventions are sensitive to bursty infectiousness. Detect-and-isolate (D&I) interventions aim to identify and isolate individuals who are or will soon be infectious, thereby limiting onward transmission. Their efficacy depends on the timing of detection relative to infectiousness.[11] When detectability is linked to infectiousness — as is realistic for symptom- or test-based detection — bursty infectiousness can cause D&I effectiveness to change dramatically with small shifts in detection time.

To quantify this, we calculated testing effectiveness (TE) — defined as the expected proportion of transmission a D&I intervention prevents [12] — for symptom-based and test-based detection as a function of ψ (**Methods**). For both detection mechanisms, TE declined as detection shifted later relative to peak infectiousness, with the sharpest drops corresponding to the narrowest infectiousness kernels (**Figure XX**). When the typical symptom onset date for SARS-CoV-2 omicron shifted from one day after to one day before peak infectiousness, this produced a xx% increase in TE at the bursty extreme ($\psi = 0$) *vs.* only a xx% increase at the smooth extreme ($\psi = 1$). Similar results held for the other benchmark pathogens and for testing-based interventions (**Figure XX, Supplementary Table XX**).

Beyond TE, D&I interventions produce two additional ψ-dependent effects. First, D&I preferentially removes late transmissions, truncating the effective generation interval distribution — an effect that vanishes as $\psi \to 0$, when detection becomes all-or-nothing and surviving transmissions proceed unimpeded. Second, bursty infectiousness amplifies the overdispersion introduced by D&I: whether detection catches an individual's narrow infectiousness burst determines whether they transmit at all, creating large variation in secondary case counts (**Supplementary Methods**). Simulations confirm that these effects shape D&I-controlled epidemic dynamics: for a given TE, bursty infectiousness produces epidemics that are less likely to establish and grow more slowly than under broad infectiousness (**Figure XX**).

Unlike D&I interventions, gathering size restrictions reduce R_{eff} (expected transmission) by a factor that is independent of ψ (**Supplementary Methods**) — but like D&I, they introduce additional ψ-dependent effects on overdispersion. By capping the size of large gatherings, gathering restrictions curtail the high-contact events that bursty infectiousness disproportionately exploits. Thus, the absolute reduction in overdispersion caused by gathering size restrictions is greatest when infectiousness is bursty (**Supplementary Methods**). Simulations confirm that gathering-size-restricted epidemics have systematically lower extinction probabilities than epidemics with the same R_{eff} but no gathering restrictions, especially for bursty infectiousness (**Figure XX**). Evaluating interventions purely in terms of ther effect on R_{eff}, without accounting for ψ, can therefore be misleading.

Burstiness complicates estimation of key epidemiological parameters. In an emerging epidemic, estimating the epidemic growth rate r and the generation interval distribution $g(\tau)$ are critical tasks, both in their own right and as ingredients for calculating R_0. Bursty infectiousness makes both tasks more challenging. First, bursty infectiousness infuses excess variation into daily case counts (**Supplementary Methods; Supplementary Figure XX**), which makes estimates of r less certain, especially when R_0 is high. Using the standard Poisson generalized linear model approach to estimate r from simulated data, we found that the standard error in $\hat{r}_{\text{MLE}}$ was between xx (for influenza) and xx (for measles) times higher for $\psi = 0$ than for $\psi = 1$ (**Figure XX**).[x]

Second, bursty infectiousness causes tight correlation among secondary infection times from a single index case. This causes the effective number of draws from $g(\tau)$ produced by an index case to collapse from R_0 when $\psi = 1$ (each secondary infection time is an independent draw from $g(\tau)$) to 1 when $\psi = 0$ (all secondary infections occur simultaneously, so together they represent a single draw from $g(\tau)$). Treating serial intervals as independent, as is common practice, can produce over-confident estimates of $g(\tau)$'s parameters (**Figure XX; Supplementary Methods**).

Discussion. The burstiness of infectiousness, as captured by the parameter ψ, plays a key role in how epidemics unfold. Just as the overdispersion parameter k illustrates how not all pathogens with the same R_0 behave alike, ψ reveals that not all pathogens with the same $g(\tau)$ behave alike. Burstiness impacts overdispersion, outbreak timing, intervention effectiveness, and epidemiological parameter estimation — a wide enough range of consequences to justify treating it as a fundamental epidemiological quantity.

Our findings have various practical implications. When designing D&I interventions, burstiness can potentially be leveraged to gain significant reductions in transmission with relatively little effort: for example, a 1-day increase in testing frequency was in some cases associated with as much as a xx% predicted increase in the intervention's effectiveness. Furthermore, burstiness can impact how interventions shape population-level transmission: accounting for an intervention's reduction in R_{eff} alone can mask important, burstiness-related shifts in epidemic growth rate and extinction probability. Since burstiness can inflate variability in daily case counts and create timing correlations among secondary infections, care must be taken when estimating and interpreting epidemiological parameter values like r and $g(\tau)$.

The ingredients for our findings have been present in the literature for over a decade, but were never assembled because standard modeling frameworks integrate over individual-level timing by construction. The individual infectiousness kernel was defined by Svensson and extended by Champredon and Dushoff. Shifts in the punctuation of the generation interval distribution have demonstrated the key impact of that population-level quantity; and replacing the stepwise infectiousness profile commonly assumed by SIR and SEIR-type compartmental models with time-varying viral kinetics-informed infectiousness kernels can generate meaningful impacts on epidemic dynamics. The course of individual infectivity has been used as key input in models of contact tracing effectiveness and testing-based interventions.

A key feature of bursty infectiousness is that it introduces an often-overlooked and unintervenable mechanism of superspreading. Coincidence superspreading produces superspreading events without superspreader individuals. This formalizes the long-observed but previously *ad hoc* distinction between superspreading events and superspreaders. As a result of coincidence superspreading, some fraction of overdispersion may be structurally untargetable, setting a floor on the overdispersion k regardless of how well we identify and manage high-risk individuals.

Our findings are subject to several limitations: the Gamma burst model, while analytically tractable, may not capture the full biological variability in burstiness. The contact models we considered are stylized, not empirically driven, and they do not account for spatial or network heterogeneity. When inferring ψ, we used serial intervals as proxies for generation intervals, inflating uncertainty in our estimates.

Looking forward, a key challenge facing future work on burstiness is measurement: estimating ψ is challenging without detailed, dense contact tracing data. Some opportunities exist for incorporating other data streams, like cross-referencing with viral kinetics; but we need a better understanding of how viral load relates to infectiousness.[3] Joint inference of k and ψ remains an open question, given that they can interact with one another. More complex models — like mixture models with multiple different values of ψ belonging to the same person — could enhance the biological fidelity of the Gamma burst model, though at the cost of model complexity.

References

[1] J. O. Lloyd-Smith, S. J. Schreiber, P. E. Kopp, and W. M. Getz. Superspreading and the effect of individual variation on disease emergence. *Nature*, 438(7066):355–359, November 2005. ISSN 0028-0836, 1476-4687. doi: 10.1038/nature04153. URL https://www.nature.com/articles/nature04153.

[2] Dylan Morris, John Maclean, and Andrew J. Black. Computation of random time-shift distributions for stochastic population models. *Journal of Mathematical Biology*, 89(3):33, September 2024. ISSN 0303-6812, 1432-1416. doi: 10.1007/s00285-024-02132-6. URL https://link.springer.com/10.1007/s00285-024-02132-6.

[3] Ashish Goyal, Daniel B Reeves, E Fabian Cardozo-Ojeda, Joshua T Schiffer, and Bryan T Mayer. Viral load and contact heterogeneity predict SARS-CoV-2 transmission and super-spreading events. *eLife*, 10:e63537, February 2021. ISSN 2050-084X. doi: 10.7554/eLife.63537. URL https://elifesciences.org/articles/63537.

[4] Centers for Disease Control and Prevention. Yellow Book: COVID-19. Technical report, Centers for Disease Control and Prevention, Atlanta, April 2025. URL https://www.cdc.gov/yellow-book/hcp/travel-associated-infections-diseases/covid-19.html.

[5] Centers for Disease Control and Prevention. About Influenza. Technical report, Atlanta, February 2026. URL https://www.cdc.gov/flu/about/index.html.

[6] Åke Svensson. A note on generation times in epidemic models. *Mathematical Biosciences*, 208(1):300–311, July 2007. ISSN 00255564. doi: 10.1016/j.mbs.2006.10.010. URL https://linkinghub.elsevier.com/retrieve/pii/S0025556406002094.

[7] David Champredon and Jonathan Dushoff. Intrinsic and realized generation intervals in infectious-disease transmission. *Proceedings of the Royal Society B: Biological Sciences*, 282(1821):20152026, December 2015. ISSN 0962-8452, 1471-2954. doi: 10.1098/rspb.2015.2026. URL https://royalsocietypublishing.org/doi/10.1098/rspb.2015.2026.

[8] Dylan J. Morris, Lauren Kennedy, and Andrew J. Black. Random time-shift approximation enables hierarchical Bayesian inference of mechanistic within-host viral dynamics models on large datasets. *PLOS Computational Biology*, 21(12):e1013775, December 2025. ISSN 1553-7358. doi: 10.1371/journal.pcbi.1013775. URL https://dx.plos.org/10.1371/journal.pcbi.1013775.

[9] Juliana C. Taube, Paige B. Miller, and John M. Drake. An open-access database of infectious disease transmission trees to explore superspreader epidemiology. *PLOS Biology*, 20(6):e3001685, June 2022. ISSN 1545-7885. doi: 10.1371/journal.pbio.3001685. URL https://dx.plos.org/10.1371/journal.pbio.3001685.

[10] Elise J. Kuylen, Andrea Torneri, Lander Willem, Pieter J. K. Libin, Steven Abrams, Pietro Coletti, Nicolas Franco, Frederik Verelst, Philippe Beutels, Jori Liesenborgs, and Niel Hens. Different forms of superspreading lead to different outcomes: Heterogeneity in infectiousness and contact behavior relevant for the case of SARS-CoV-2. *PLOS Computational Biology*, 18(8):e1009980, August 2022. ISSN 1553-7358. doi: 10.1371/journal.pcbi.1009980. URL https://dx.plos.org/10.1371/journal.pcbi.1009980.

[11] Christophe Fraser, Steven Riley, Roy M. Anderson, and Neil M. Ferguson. Factors that make an infectious disease outbreak controllable. *Proceedings of the National Academy of Sciences*, 101(16):6146–6151, April 2004. ISSN 0027-8424, 1091-6490. doi: 10.1073/pnas.0307506101. URL https://pnas.org/doi/full/10.1073/pnas.0307506101.

[12] Casey Middleton and Daniel B. Larremore. Modeling the transmission mitigation impact of testing for infectious diseases. *Science Advances*, 10(24):eadk5108, June 2024. ISSN 2375-2548. doi: 10.1126/sciadv.adk5108. URL https://www.science.org/doi/10.1126/sciadv.adk5108.

Figure Legends

Figure 1 | **The individual infectiousness kernel and the burstiness parameter ψ.**

Figure 2 | **Empirical estimates of ψ.**

Figure 3 | **Coincidence overdispersion.**

Figure 4 | **The epidemic latent period.**

Figure 5 | **Interventions.**